

More efficient eventstudies

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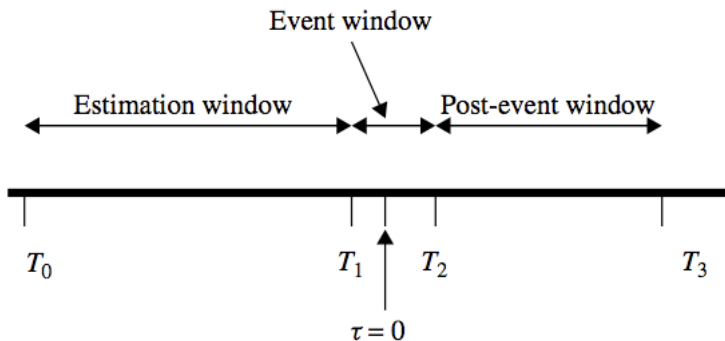
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Measuring effects

- Economists are always asked to measure effects of economic events.
- Event studies is the broad framework that attempts to understand effects due to a specific event / set of events.
- Long history. Dolley (1933) examined price effects of stock splits.
- Popularised by Ball and Brown (1968), Fama, Fisher, Jensen and Roll (1969), Brown and Warner (1980, 1985).
- Renewed interest today with causal inference gaining momentum.

The framework



A naïve event study

- Let 'day-0' identify the event in the event-time τ .
- Let $\tau = \dots, -3, -2, -1$ represent trading days leading up to the event.
- A naïve event study will:-
 - 1 Compare event date return R_0 with past performance.
 - 2 Returns observed during a control period before the event.
- However, several problems exist with a direct comparison:-
 - 1 Event-time contamination
 - 2 Event contamination - other events

A more sophisticated approach: Market models

- Market models - work horse of finance literature.
- A market model for security i and observation τ in event time is:-

$$r_{i,\tau} = \alpha_i + \beta_i r_{m,\tau} + \epsilon_{i\tau}$$

- Under general conditions, the OLS is a consistent, efficient estimation procedure for the market model parameters.
- Problem: CAPM just doesn't work.

More controls in a linear world: Factor models

- An array of factor models from literature. E.g: Fama and French factors.
- A linear factor model for security i and observation τ in event time is:-

$$r_{i,\tau} = \alpha_i + \sum_{k=1}^N \beta_i F_{k,\tau} + \epsilon_{i,\tau}$$

- $F_1, \dots, F_N$ factors.

Statistical inference

- The standard deviation of the average abnormal return will be:-

$$\sqrt{\frac{\sum_{t=T_0}^{T_1} \{AR_{it} - \bar{AR}_{it}\}}{T_1 - d}}$$

- Under the null of no abnormal performance, the statistic above is distributed as Student-t with $T - d$ degrees of freedom.
- *To adjust for cross sectional dependence* is to estimate the standard deviation over the estimation period (Brown and Warner (1980)).

Statistical power

- Statistical power of the analysis depends on the attempt to explain all but the abnormal returns associated with the event.
- With more variables, it is likely that we explain more of the returns.
- Factor models and other augmented market models achieve this reduction
- *Key question:* Can we achieve further increase in non-idiosyncratic variance for a given set of event stocks?

The ideal setting

- ① A counterfactual to the event stock
- ② What would the firm look like had the event not taken place?

Achieves the best reduction in idiosyncratic share of the returns variance, thus statistical power.

Capturing non-linearities

- Models that capture non-linearities, is one step of the way.
- Such non-linearities could be achieved using polynomials of available suite of risk factors.
- However, a tailor-made counterfactual based on a suite of firm characteristics (e.g: market cap, price to book) captures the nonlinearities better.
- Suppose $CF_{i,\tau}$ is the counterfactual.

$$r_{i,\tau} = \alpha_i + \beta_{i,CF} CF_{i,\tau} + \epsilon_{i\tau}$$

- Better CF s will make $\beta_{i,CF} \rightarrow 1$.

A new approach: Mahalanobix

- Multivariate-distance measurement between event stock and other stocks in the market:

$$\sqrt{(x - \mu)^T S^{-1} (x - \mu)}$$

covariance matrix

- Intuition: If S^{-1} is diagonal, then we obtain normalized Euclidean distances.
- Allows for a tailor-made counterfactual to every event stock (CF):
 - 1 Equally-weighted index return
 - 2 Inverse-distance weighted index return
- A non-parametric approach to capturing non-linearities of the risk factors / firm characteristics

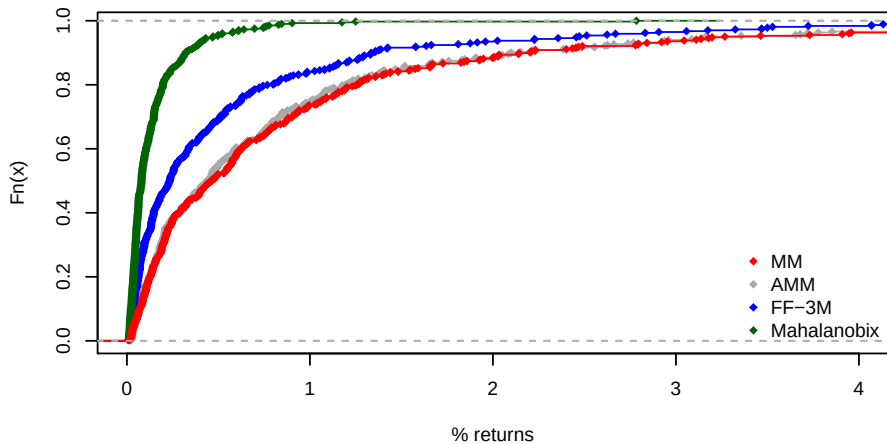
Horse race to power

- Statistical power: Percentage of correct rejection of H_0 .
- Simulate 100 times with varied sample size and shock size for each of the following six models of adjustment
 - 1 Excess returns to the market (ER)
 - 2 Market model (MM)
 - 3 Augmented market model (with currency risk) (AMM)
 - 4 Fama-French 3 factor model (FF-3M)
 - 5 Mahalanobis - equal weights and inverse-distance weights (M1, M2)
- *ex-ante* expectation:
 - 1 Variance: (broadly) monotonic reduction
 - 2 Power: (broadly) monotonic increase

Data

- Universe of listed firms in India (2008 - 2014)
- Daily closing price returns, currency and market returns: CMIE Prowess
- Fama-French factors from Jayanth Varma's data library

Variance reduction



Power - Percentage of correct rejections of null

Shock size: 1%

Sample size	MM	AMM	FF-3M	M1	M2
5	23.00	26.53	26.53		
10	24.49	33.33	28.89		
15	32.35	50.00	56.25		
20	44.00	62.50	59.32		
25	47.37	73.33	63.64		

Power of the test

Shock size: 2%

Sample size	MM	AMM	FF-3M	M1	M2
5	20.00	24.49	26.26		
10	27.27	38.30	41.94		
15	45.83	66.15	53.85		
20	60.47	78.38	63.89		
25	73.58	80.43	76.74		

Power of the test

Shock size: 3%

Sample size	MM	AMM	FF-3M	M1	M2
5	16.33	28.42	25.26		
10	44.00	52.13	55.43		
15	56.34	73.85	59.09		
20	73.91	78.95	66.67		
25	84.21	81.25	80.00		

Next steps

- Case study: When there is one event
- Expand on the suite of statistical tests for power study
- Synthetic returns: Counterfactuals for very large firms
- Power in long-horizon event studies (notorious for low power in the literature)

Thank you